

A study using the FWPA softwood data series

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Summary

This report was jointly funded by Forest and Wood Products Australia (FWPA) and ABARES to estimate the responsiveness of demand for structural pine to changes in timber and steel prices. Measures of demand responsiveness can provide valuable insights into the potential implications of changes in policy or market settings on volumes and prices received by producers. The analysis focuses on estimating short-term price elasticities of demand—a formal measure of the sensitivity of demand—to changes in prices in the same quarter or recent past.

For this report ABARES estimated price elasticities of demand for domestically produced structural pine (FWPA softwood data series) with respect to the domestic price of MGP10 (machine graded pine), imported structural pine prices, and prices for steel beams and sections used in housing construction.

ABARES tested a large number of models with varying functional forms and combinations of variables to estimate price elasticities of demand. The estimated models of demand are based on historical relationships between the FWPA softwood sales data series and various explanatory variables. The estimates outlined in this report should also be considered in the context of the assumptions and limitations of the data used in ABARES modelling. In particular, the models of demand developed by ABARES are considered to be a proxy for national demand, as they only cover producers who are included in the FWPA survey.

The estimates presented in this report are based on industry-level demand and average prices. As such, they are not applicable to changes in demand or prices of individual producers. Changes in prices of individual producers will likely lead to much larger changes in demand for their products, but little to no change in industry-level demand.

The estimates suggest timber and steel prices are not major determinants of demand for domestically produced structural pine products in the short term, with non-price factors playing a more important role. The report presents insights into these non-price factors—such as house commencements, environmental concerns, consumer preferences and building code changes—as alternative determinants of demand. The findings presented in this report are broadly consistent with previous estimates of price elasticities of demand for structural timber in Australia and internationally.

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Key findings

Timber and steel prices

Changes in the domestic price of structural pine (MGP10) have a relatively small impact on demand for domestically produced structural pine (Table S1). A 1 per cent increase in MGP10 prices was estimated to reduce demand for domestically produced untreated structural pine by between 0.06 and 1.28 per cent, reduce demand for treated structural pine by between 0.54 and 0.82 per cent and reduce demand for the aggregate of the two by between 0.08 and 0.55 per cent. However, a number of these estimates were not statistically different from zero at the 95 per cent confidence level, suggesting potentially no effect on demand for structural pine. There was evidence that changes in domestic prices were more likely to have a delayed, rather than immediate, impact on demand—with changes in prices in the previous quarters having a statistically significant effect on demand.

Changes in import prices for structural pine products had a small effect on demand for domestically produced structural pine. A 1 per cent increase in imported structural pine prices was estimated to increase demand for domestically produced untreated structural pine by up to 0.14 per cent, increase demand for domestically produced treated structural pine by up to 0.18 per cent, and increase demand for the aggregate of the two by up to 0.20 per cent. These estimates were statistically insignificant at the 95 per cent confidence level, which implies that import prices may have little effect on demand for domestically produced structural pine in the short-term.

Table S1 Change in demand for domestically produced structural pine in response to 1 per cent change in timber and steel prices

Price variables	Untreated structural pine (%)	Treated structural pine (%)	Aggregate structural pine (%)	Domestic share (%)
Domestic MGP10 price	−1.28 to −0.06 b	−0.82 to −0.54 b	−0.55 to −0.08 b	−0.42 to − 0.21 b
Imported structural pine prices	−0.13 to 0.14	−0.03 to 0.18	−0.01 to 0.20	−0.06 to 0.30 b
Price of steel beams and sections used in housing—whole period	−0.36 to −0.09	−0.54 to 0.18 b	−0.35 to 0.15	0.08
Price of steel beams and sections used in housing—before September quarter, 2007	1.27 a	—	0.99 a	—
	−0.16	—	−0.44	—

Price of steel beams and sections used
in housing—during and after
September quarter, 2007

a Estimate statistically significant at the 95 per cent confidence level. b Some estimates in this range were statistically significant at the 95 per cent confidence level.

Note: There was no evidence of a structural break with respect to steel prices for models of treated structural pine or the domestic market share.

However, there is evidence that changes in import prices may have an impact on the share of domestic demand met by domestic production. For example, a 1 per cent increase in import prices was estimated to increase the market share of domestically produced structural pine by up to 0.30 per cent, with a number of the estimates being statistically significant. These results are consistent with changes in import prices affecting demand for imports to a greater degree than demand for domestically produced structural pine.

The price of steel beams and sections used in housing construction was estimated to have a negative effect on demand for domestically produced structural pine over the sample period. That is, increases in steel prices reduced demand for domestically produced structural pine. This finding is inconsistent with steel and timber being price substitutes. However, when the responsiveness of demand to steel prices was allowed to change over the sample period, the results indicated strong and statistically significant price substitution between steel and domestically produced structural pine before the September quarter of 2007, or the beginning of the global financial crisis (GFC). During and after the GFC, however, steel prices appeared to have little effect on demand for domestically produced structural pine.

It should be noted, however, that the ABS price index for steel beams and sections used in housing is an aggregate of multiple steel products, some of which may not be directly substitutable with structural pine. This could, in part, explain why some of the model results indicate a limited effect of steel prices on demand for domestically produced structural pine after the September quarter of 2007.

Overall, ABARES estimates suggest a weak relationship between timber and steel prices and demand for domestically produced structural pine—many of the estimated coefficients are statistically insignificant.

Non-price factors

Changes in the number of house commencements and, to a lesser degree, the value of work done on new houses have a more substantial impact than prices on demand for domestically produced structural pine (Table S2). A 1 per cent increase in new house commencements was estimated to increase demand for domestically produced untreated pine by up to 0.64 per cent, demand for domestically produced treated pine by up to 0.99 per cent, and demand for the aggregate of the two by up to 0.69 per cent. The effect of house commencements on structural timber demand was found to be statistically significant in all models considered but the effects of the value of work done was mixed.

Table S2 Change in demand for domestically produced structural pine in response to 1 per cent change in residential housing activity

Variables	Untreated structural pine (%)	Treated structural pine (%)	Aggregate pine (%)
House commencements a	0.36 to 0.64	0.77 to 0.99	0.48 to 0.69
Value of work done on new houses	0.14 to 0.31	0.00 to 0.33	0.26 to 0.44 b

a All estimates were statistically significant at the 95 per cent confidence level. b Some estimates in this range were statistically significant at the 95 per cent confidence level.

Other non-price factors may also explain why changes in the relative prices of timber and steel are not necessarily matched by a corresponding change in consumption. For example, differences in order lead times between materials can offset any cost savings associated with lower material prices. Long order lead times impose costs on builders—including idling capacity, having to turn down potential projects and difficulty planning for the future. Offsite prefabrication (of timber frames, for example) can reduce build times and onsite flexibility can minimise delays. These have implications for build times and labour costs that aren’t accounted for in material prices alone. When these additional factors are taken into account, a large price differential may be required before substitution between materials occurs.

Changing consumer preferences has also had an effect on the choice of construction material. Consumers are increasingly opting for open plan houses, which require fewer internal walls, decreasing the required amount of material inputs. However, this means structural construction systems need to be stronger, placing a greater emphasis on strength and durability of the material used. With environmental issues becoming increasingly important, consumers may place more weight on the environmental properties of various construction systems and materials. Recent changes to the National Construction Code could allow for increased timber use in the midrise construction market, leading to potentially higher timber use in multi-storey buildings in the future.

How to read this report

Chapter 1 provides a brief background on ‘price elasticities of demand’ as a measure of demand responsiveness and discusses the key datasets used. Chapter 2 presents estimates of price elasticities of demand with respect to timber and steel prices and Chapter 3 discusses non-price factors which play an important role in determining demand for domestically produced structural pine. The report concludes with a brief discussion of key results and implications. Appendix A discusses the technical aspects of model development while Appendix B presents detailed model outputs.

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